Synthesis of Pentyl Caprylate via Fischer Esterification

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1 Introduction

The objective of this experiment is to synthesize pentyl caprylate via Fischer Esterification by reacting absolute 1-pentanol with caprylic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

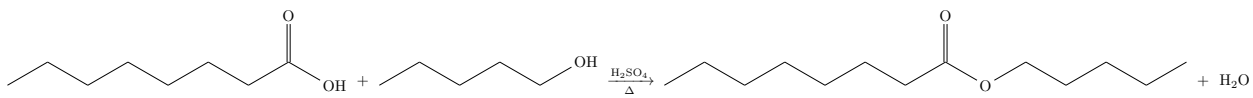


Figure 1: Reaction of Caprylic Acid and 1-Pentanol

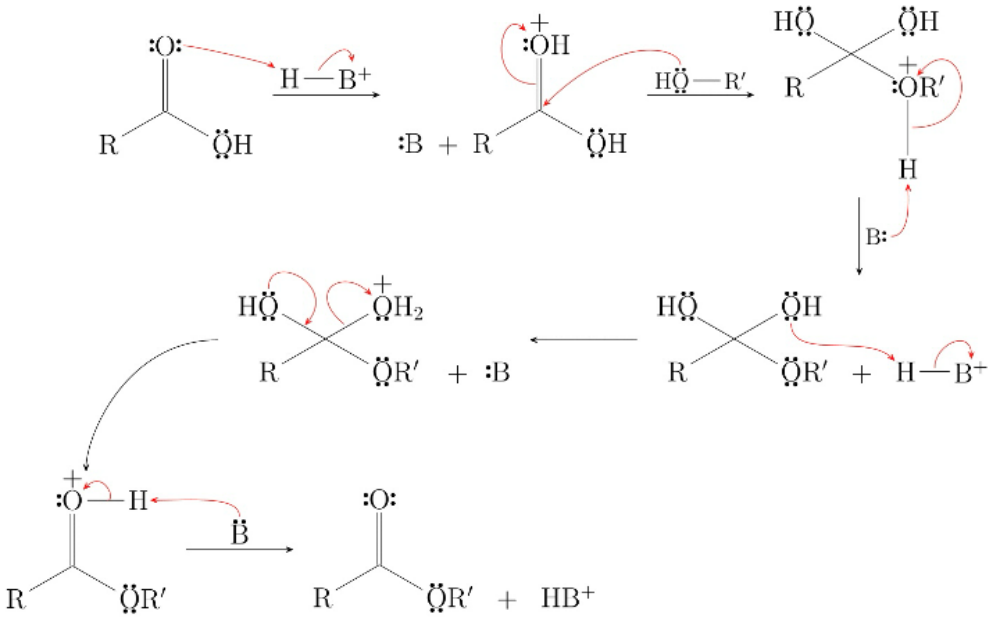


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Caprylic Acid	144.21	0.91	237 – 239.7	1.0	5.786 g / 6.36 mL	40.1
1-Pentanol	88.15	0.811	137 – 139	2.0	7.0 g / 8.7 mL	80.2
Sulfuric Acid	98.08	1.84	–	cat.	5 drops	–
Diethyl Ether	74.12	0.710	34.6	solv.	–	–
Pentyl Caprylate	214.35	0.86	262	–	8.60 g / 10.0 mL	40.1

4 Experimental Procedure

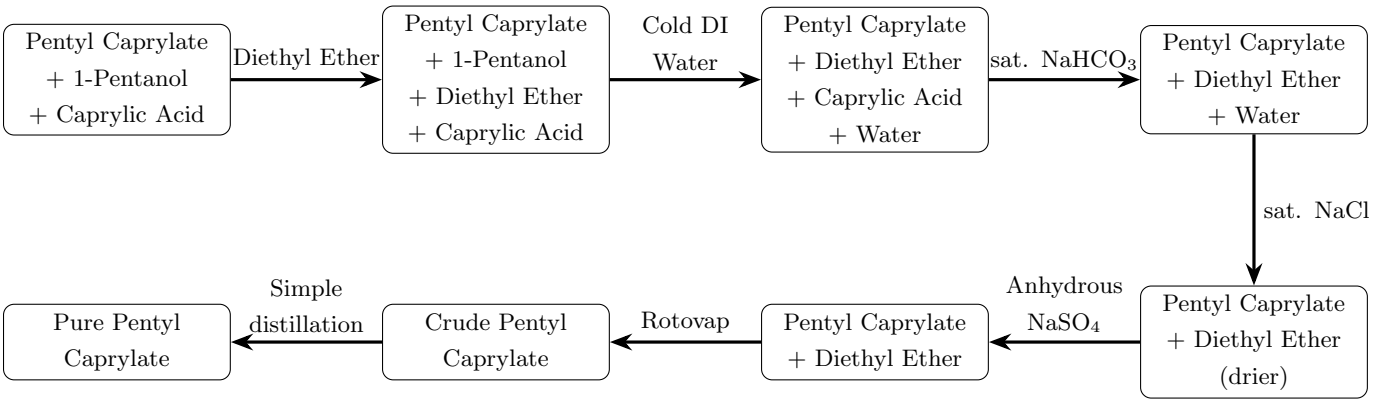
4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

4.2 Procedure

- In a 25 mL round-bottom flask, add 5.786 g (6.36 mL) of caprylic acid followed by 7.0 g (8.7 mL) of 1-pentanol and a stir bar.
- While stirring, add 5 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 15 mL total of diethyl ether to aid in the process.
- Quench the organic layer with iced cold DI water and collect the organic layer. Do this twice.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude pentyl caprylate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Caprylic acid	g of 1-Pentanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations